



Group 11 Presentation

MSCI151 Group Methex 1

Introduction



Determine optimal ordering policies for 3 different scenarios for a shop



Set up simulations for each part using excel



Calculated optimal values and then tested values around these values



Drew conclusions based on results

The Simulation Model

- The simulation model for both the Sprint and Vitality drinks include the day, inventory at the start of the day, any deliveries arriving for the day using the formula **COUNTIF(\$M\$4:M4,A5)*\$Q\$13** to check in the day received column whether any deliveries are arriving on the day, and how many units will be arriving
- Then there is the available stock, which is the inventory at the start of the day plus the deliveries (if any are arriving on the day)
- Then we added the demand and demand satisfied column using the formula **MIN(D5,E5)** to see whether the demand is lower than the stock and if all order have been fulfilled for the day

The Simulation Model

- The next column is end inventory with the formula $\text{MAX}(D5-E5, 0)$, and this calculates the inventory at the end of the day once all orders have been fulfilled
- The shortage column is next using the formula $\text{MAX}(E5-D5, 0)$ which calculates if there were any shortages and how many by subtracting the demand from the available stock
- The formula used for the stockout column is $\text{IF}(H5>0, 1, 0)$ and this calculates whether the available demand was enough to satisfy the demand

The Simulation Model

- The formula for the end inventory + order column is $J4 - F5 + IF(K4=1, \$Q\$13, 0)$ and this calculates the inventory at the end of the day as well as adding the orders awaiting arrival
- The next column is the order column, and it determines whether a new order should be placed according to the stock in the end inventory and order column, and whether it is lower than the re-order level or not. It uses the formula $IF(J5 \leq \$Q\$12, 1, 0)$
- The next column is the lead time column which determines how many days the delivery will take if one is placed, using probability and random distribution, and the formula $IF(K5=1, LOOKUP(RAND(), \$R\$4:\$R\$5, \$P\$4:\$P\$5), 0)$. (A bit more on this later)

The Simulation Model

- The next column is the day received column which provides the day on which the delivery will arrive if one is placed, and uses the formula $\text{IF}(K5=1, L5+A5+1, \text{"N/A"})$ to add the lead time to the current date if an order is placed
- The final column is the cost column which calculates the total cost for the day using the formula $(G5*\$Q\$14) + \text{IF}(K5=1, \$Q\$15+(\$Q\$13*\$Q\$16), 0) + (H5*\$Q\$16)$ This adds the holding cost of per unit for however many units are in the end inventory, the stockout cost (if applicable), and if any order were placed it adds the order cost of £30 as well as the order cost per unit

The Simulation Model

- Another important point is that all simulations had 100 replications performed
- To use these values a so called 'average average daily cost' was used to ensure that the results were accurate

The Simulation Model

	Day	Inventory (start)	Delivery	Available stock	Demand	Demand satisfied	Inventory (end)	Shortage	Stockout? Y/N	End inventory + Order	Order? Y/N	Lead time	Day received	Cost		Lead time									
	01/01/2022	100	0	100	25	25	75	0	0	75	1	4	06/01/2022	£147.15			4	0.9	0				Demand SD	5.03597914	
	02/01/2022	75	0	75	27	27	48	0	0	237	0	0 N/A		£2.40			5	0.1	0.9				Mean	29.6565934	
	03/01/2022	48	0	48	23	23	25	0	0	214	0	0 N/A		£1.25									Confidence Level	0.95	
	04/01/2022	25	0	25	28	25	0	3	1	189	0	0 N/A		£3.60									Z Score	1.64485363	
	05/01/2022	0	0	0	33	0	0	33	1	189	0	0 N/A		£39.60									E(Lead Time)	4.1	
	06/01/2022	0	189	189	26	26	163	0	0	163	0	0 N/A		£8.15									Safety Stock	17	
0	07/01/2022	163	0	163	33	33	130	0	0	130	1	4	12/01/2022	£149.90									Lead Time demand	122	
1	08/01/2022	130	0	130	36	36	94	0	0	283	0	0 N/A		£4.70									Reorder Level	138	
2	09/01/2022	94	0	94	31	31	63	0	0	252	0	0 N/A		£3.15		Re-order level		138					Order Quantity	189	
3	10/01/2022	63	0	63	30	30	33	0	0	222	0	0 N/A		£1.65		Order quantity		189							
4	11/01/2022	33	0	33	28	28	5	0	0	194	0	0 N/A		£0.25		Holding cost		£0.05							
5	12/01/2022	5	189	194	32	32	162	0	0	162	0	0 N/A		£8.10		Order cost		£30.00							
5	13/01/2022	162	0	162	29	29	133	0	0	133	1	4	18/01/2022	£150.05		Stockout cost		£1.20							
7	14/01/2022	133	0	133	34	34	99	0	0	288	0	0 N/A		£4.95		Order cost per unit		£0.60							
7	15/01/2022	99	0	99	25	25	74	0	0	263	0	0 N/A		£3.70											
9	16/01/2022	74	0	74	31	31	43	0	0	232	0	0 N/A		£2.15											
0	17/01/2022	43	0	43	31	31	12	0	0	201	0	0 N/A		£0.60											
1	18/01/2022	12	189	201	33	33	168	0	0	168	0	0 N/A		£8.40		Average demand		30							
2	19/01/2022	168	0	168	25	25	143	0	0	143	0	0 N/A		£7.15		Number of stockouts		21							
3	20/01/2022	143	0	143	24	24	119	0	0	119	1	4	25/01/2022	£149.35		Average shortage		17							
4	21/01/2022	119	0	119	27	27	92	0	0	281	0	0 N/A		£4.60		Percentage of demand satisfied		96.76%							
5	22/01/2022	92	0	92	28	28	64	0	0	253	0	0 N/A		£3.20		Average cost per day		£27.53							
5	23/01/2022	64	0	64	22	22	42	0	0	231	0	0 N/A		£2.10		Average average cost per day		£27.46							
7	24/01/2022	42	0	42	30	30	12	0	0	201	0	0 N/A													

[illegible]

Calculating Lead Time

- Looking at Sprint, however the same method applied to the rest of the simulations
- The lead time for the Sprint drink was calculated using the given probabilities of 90% for a lead time of 4 days and a 10% chance of a lead time of 5 days
- We created a table using these probabilities and used the **RAND()** function in the lead time column to generate a random number between 0 and 1
- We then used the **LOOKUP** function refer back to the table created, and values between 0 and 0.89 would give a lead time of 4 days and values between 0.9 and 0.99 would give a lead time of 5 days

Ordering Policies – First Approach

- Trial and error was not an efficient way to find optimal values for reorder levels and quantity due to the wide range of possibilities with two variables
- Therefore, we calculated theoretically the optimal values for Order Quantity and Reorder Level

- Economic Order Quantity =
$$\sqrt{\frac{2 * \text{Fixed Order Cost} * \text{Average Demand}}{\text{Holding Cost per Unit}}}$$

- Reorder Level = $\text{Safety Stock} + \text{Lead Time Demand}$

- Safety Stock = $\sqrt{E(\text{Lead Time})} * Z_{\text{confidence}} * \sigma_{\text{demand}}$

- Lead Time Demand = $E(\text{Lead Time}) * \text{Average Demand}$

Ordering Policies – Second Approach

- Applied skills from the workshop to use a solver to find the optimal reorder and quantity levels for both Sprint and Vitality respectively
- Objective of the solver is set to minimize the value of cell which is summation of daily costs
- The variable cells changed are the order quantity and reorder levels
- Added constraint for order quantity and reorder levels to be integers to be a suitable solution
- GRG Nonlinear solving method used as this was used in workshop to obtain optimal values
- Interestingly, our calculated values were very close to the solvers' optimal values. For Vitality, they were the same

Ordering Policies – Second Approach

Draw Page Layout Formulas Data Review View Help Acrobat

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

\$Q\$12 = integer

\$Q\$13 = integer

Add

Change

Delete

Reset All

Load/Save

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

27								
141								
117								
113								
97								
77								
67								
46								
29								
18								
0								
110								
95								
78								
66								
52								
46								
28								
9								
124								
109	0	89	18	18	71	0	0	
89	0	89	18	18	71	0	0	
71	0	71	17	17	54	0	0	

Ordering Policies – Final Values

- Once we had the solver's optimal values for the Average Daily Cost, we then played around with the number manually to try and then minimise the number of stockouts low and maximise the percentage of demand satisfied as high as possible.
- This led to some slight changes in values, but we decided not to stray too far from the Average Daily Cost optimal value as this was what the store was primarily interested in
- The exercise could be run again to focus more on these other factors should the store require it

Conclusions – Part A

- Concluded that the optimal values were:
 - Sprint
 - Reorder Level = 151
 - Order Quantity = 185
 - Average Daily Costs = £27.71
 - Vitality
 - Reorder Level = 66
 - Order Quantity = 130
 - Average Daily Costs = £21.73

Conclusions – Part B

- Concluded that the optimal values were:
 - Sprint
 - Reorder Level = 74
 - Order Quantity = 218
 - Average Daily Costs = £28.53
- Despite a slightly higher average cost, we would recommend the store to order from Supplier C as the number of stockouts were reduced.

Conclusions – Part C

- Using the optimal values discovered in Part A, the model was applied with policy being more heavily thought about.
- The best policy was as follows:
 - When one drink falls below its reorder level, an order is placed for the reorder quantity of that drink
 - Then reorder the other drink at a quantity equal to
 - $\text{Standard Order Quantity} - (\text{Stock Level} - \text{Reorder Level})$
- This minimises the number of orders hence minimising the number of orders required hence minimising fixed order cost.

Part C Example

- With Standard Values:

	Sprint	Vitality
Order Quantity	185	130
Reorder Level	151	66

Day	Sprint Stock	Vitality Stock	Reorder Needed (S/V)	Order Quantity S	Order Quantity V
01/06/2022	210	79	No	0	0
02/06/2022	190	57	Yes (Vitality)	=185-(190-151) =146	66

Recommendations

- From parts A, our final recommendation to the store would be to purchase from Supplier A and B with the L_s , L_v , Q_s , Q_v values given previously. Should supplier C be available to order from as in Part B, we would suggest that the store does so.
- Should suppliers A and B merge as they do in part C, our recommendation would be to order on a model as given, by adjusting the order levels to suit the current inventory level.
- Obviously, all of these recommendations are based on a previous years' data so the store needs to be aware that demand may vary.



Any questions?